

SatarkPay

Paisa bhejne se pehle 60 second. Fraud ke baad 60 minute.

SANGYAN — Investor Resilience Hackathon 2026

SEBI × NSDL × SNTCL, IIT (BHU) Varanasi · Track A + B · Team SCAMURAI

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Live demo: sudonishant.github.io/satarkpay-repo

Code + tests: github.com/sudonishant/satarkpay-repo · MIT · 66/66 checks · offline chalta hai

Ramesh ke saath aaj subah kya hua

9:41

WhatsApp aaya — “HR Priya”,
₹5,000/din ka kaam offer

9:47

“Registration fee ₹499” —
UPI link bheja

9:53

Ramesh payment screen par
pahunch gaya

9:54

Paisa gaya. Number block.
Paisa wapas? Pata nahi.

Sirf 13 minute. Us dauran Ramesh ke paas koi tool nahi tha jo beech me aa jaata.

Aaj ke tools payment ke BAAD aate hain — score padhne se paisa wapas nahi aata.

SatarkPay us 13 minute ke andar kaam karta hai — aur uske baad ke 60 minute bhi sambhalta hai.

Ye Ramesh akele nahi hain

16 crore+

demat accounts — 70% naye
Tier-2/3 cities se

9 / 10

individual F&O traders
net loss me (SEBI study)

71%

competitor repos sirf
“paste → score” checker

24%

ke paas koi working
live demo hi nahi

Aur aaj ki tools fraud ke BAAD aati hain.

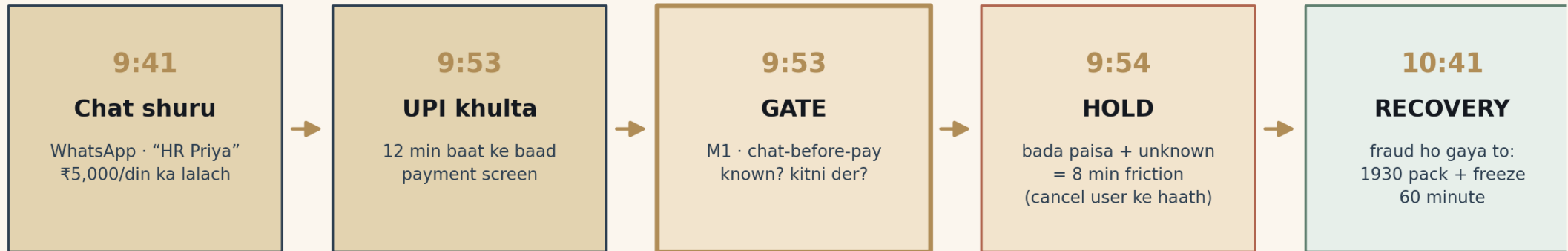
User message paste karta hai → score milta hai → par paisa ja chuka hota hai. Recovery ke liye phir bhi koi plan nahi.

Isliye humne do cheezein banayi: PAISA ROKNA (pre-pay) aur PAISA WAPAS (evidence pack).

Chat se payment tak — beech me ek guard

Ek asli journey — chat se payment tak, aur fraud ke baad tak

SatarkPay chat ke andar nahi jhaankta — sirf timing + payment signals padhta hai (on-device)



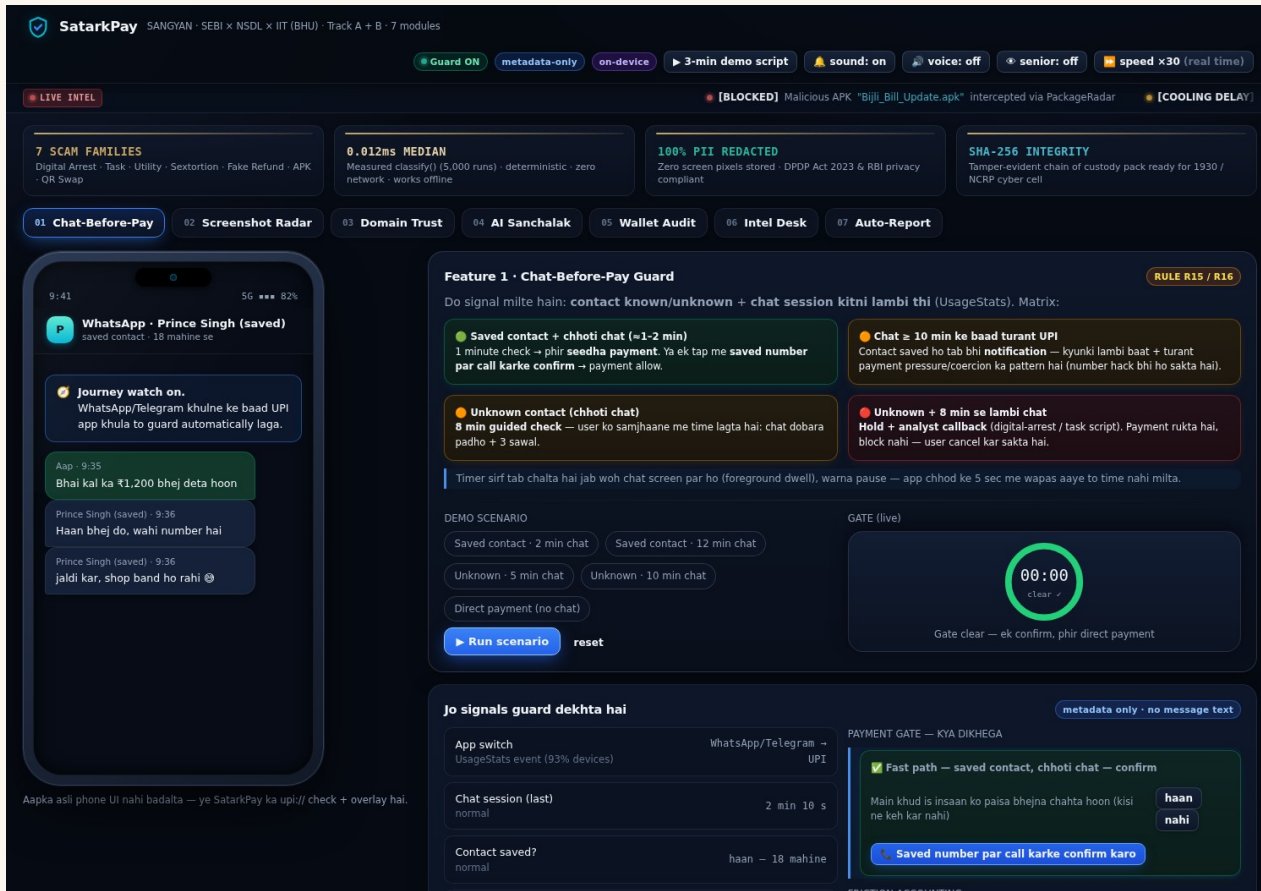
60 SECOND · paisa jaane se pehle

60 MINUTE · fraud ke baad

Kya khaas hai:

- Guard chat ke andar nahi jhaankta — sirf timing + payment signals (on-device, 0 message text).
- Friction hazaar logon par nahi — sirf us combo par jo fraud ka shape hai (unknown + lambi chat + bada amount).
- Block kuch bhi auto nahi — hold aata hai, final call user ki. (Ethics + guardrail compliance)

Judge khud chala sakta hai — 2 minute me



M1 · Chat-Before-Pay

saved/unknown × chat length ka matrix

M2 · Screenshot Radar

payment screenshots ka burst + provenance

M3 · Domain Trust

gateway ≠ merchant mismatch

M4 · AI Sanchalak

verdict + registry format + negation-aware

M5/M6/M7 · audit · intel · recovery

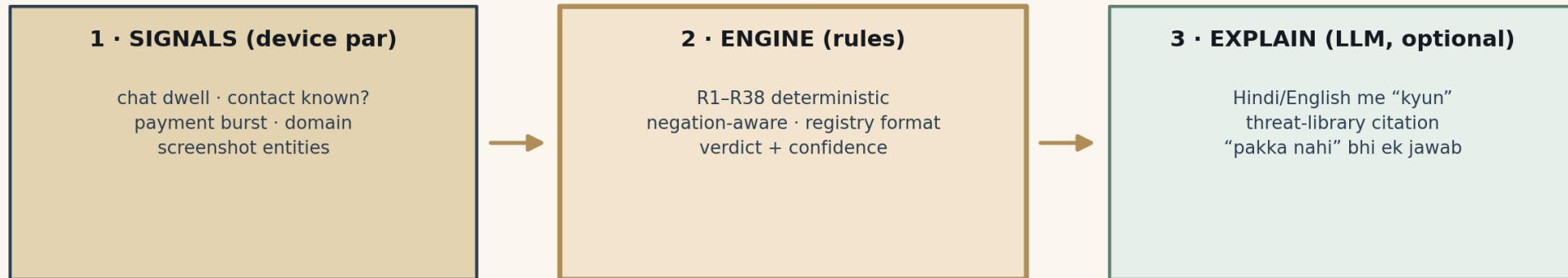
emergency layer: email + 1930 + payment stop

Live: sudonishant.github.io/satarkpay-repo

offline bhi chalta hai · 164 KB single file · koi install nahi

Verdict rules dete hain — LLM sirf samjhata hai

Verdict deterministic hai — LLM sirf samjhata hai (kyun), decide nahi karta



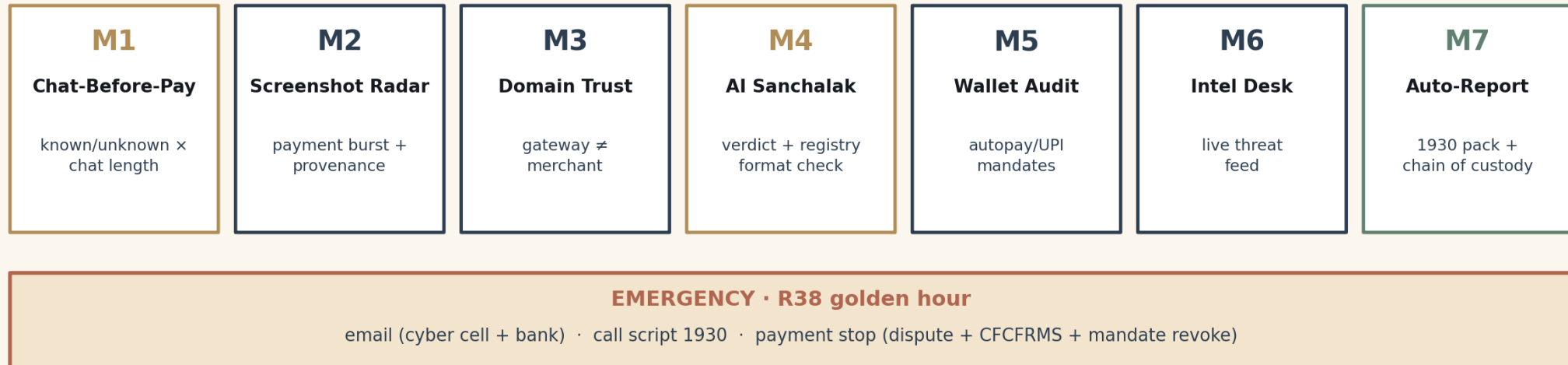
Network zero · inference 0.012 ms (median, 5,000 runs) · offline demo 164 KB single file

Isliye hum bharosa karne layak hain:

- Deterministic engine = same input, same verdict (audit ho sakta hai) · LLM off ho to bhi system chalta hai
- “Pakka nahi bata sakta” ek valid answer hai — aur eval me hum apne miss/FP chhupate nahi

Har module ek alag scam pattern pakadta hai

Saat modules + emergency layer — poora chain: pehle roko, baad me likho



Emergency layer sirf tabs par nahi — M1 ke HOLD card se bhi ek tap me khulta hai.

Emergency layer (R38) sirf ek tab nahi — M1 ke HOLD card, M7 ke report aur panic button se ek tap me khulta hai.

Demo me sab kuch offline chalta hai: koi API key, koi network, koi install nahi.

33 messages · P 95% · R 90.5% — aur misses bhi dikhte hain

33 labelled messages — aur hum apne miss/FP khud dikhte hain

Metrics



Confusion (33 msgs)

19 TP	1 FP
2 miss (FN)	9 TN

“PAKKA NAHI BATA SAKTA” ek valid jawab hai — isliye 9 legit messages ko humne chhoda, guess nahi kiya.

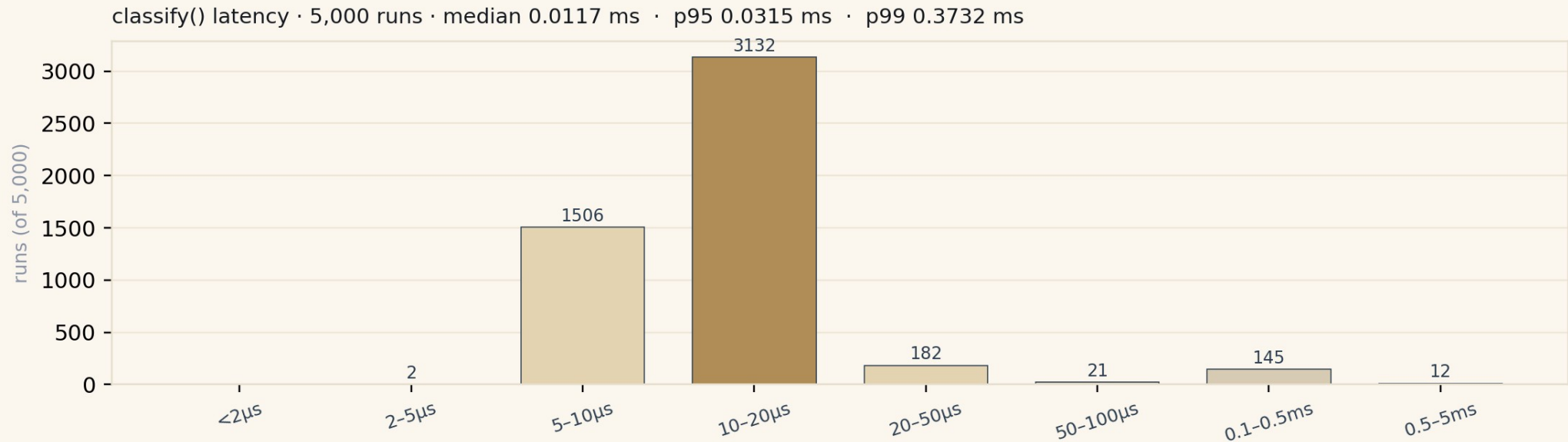
Set self-authored hai (ceiling, field accuracy nahi) — ye hum likhte hain, chhupate nahi.

Command: `python3 eval/run_eval.py · latest_results.json` repo me

Self-authored set ka matlab: ye CEILING hai, field accuracy nahi — hum yahi likhte hain, chhupate nahi.

Miss 2 aur false-alarm 1 — teeno messages ka pura text repo me hai, judge khud verify kar sakta hai.

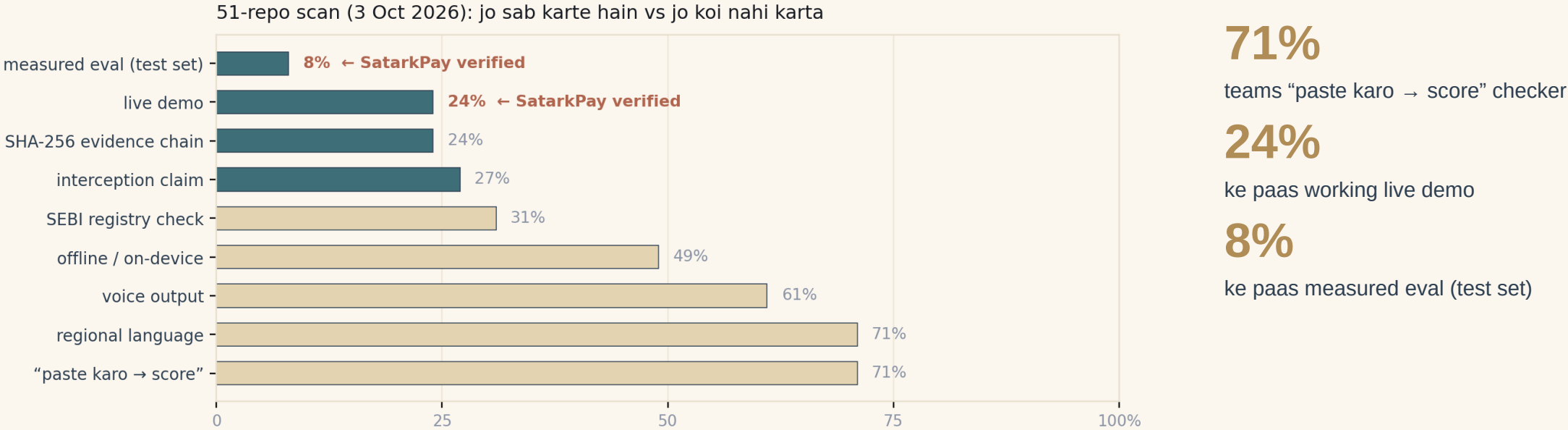
0.012 ms median — 5,000 runs ka asli histogram



Ye chart raw data se bana hai (latency_raw.json) — hand-drawn nahi. Command: eval/bench_latency.js

Network call zero — “on-device” ka yahi matlab. Low-end phone par zyada hoga; wo re-measure Phase-2 me hai (honest note).

Jo sab bana rahe hain, hum wo nahi banaye



SatarkPay = interception + evidence-grade recovery + measured proof — teeno ek saath kisi ke paas nahi.

Hum wo permissions maangte hi nahi, jo scam apps maangte hain

Guardrail: hum wo permissions maangte hi nahi jo fraud apps maangte hain

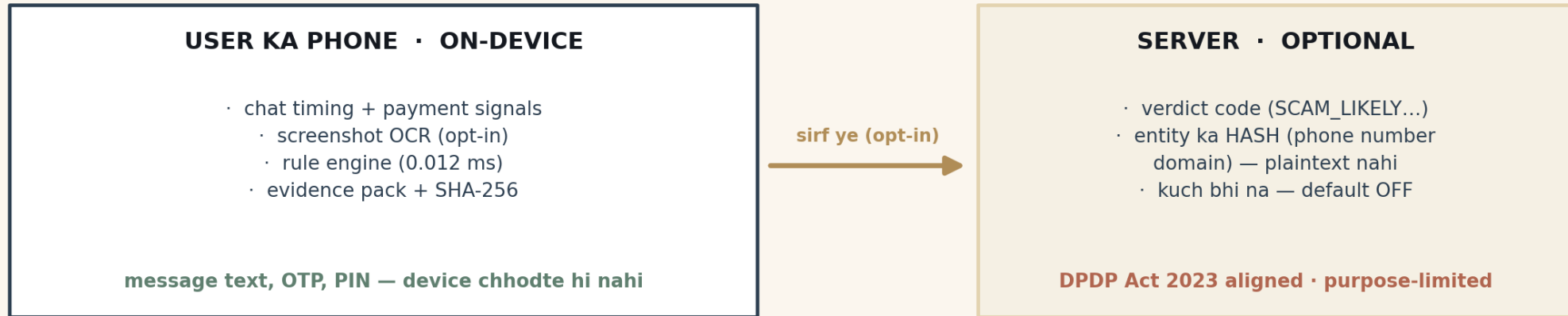
	SatarkPay (manifest verified)	industry pattern
READ_SMS / SMS padhna	X SatarkPay	✓ typical scam apps
Notification access	X SatarkPay	✓ “wallet audit” apps
Contacts / Gallery	X SatarkPay	✓ “convenience” apps
Internet + audio	✓ sirf ye 2 (+2 harmless)	✓✓✓

PS ke guardrails ke andar, sab likhit:

koi stock tip / buy-sell-hold nahi · koi monetisation nahi · SMS/OTP/PII harvest nahi · uncertainty transparent · public-good ethos
Privacy by design: DPDP-aligned · data device par · server par sirf verdict code + entity hash (opt-in).

Data kahan jaata hai — phone se bahar kya nikalta hai

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Judges note: yahi wajah hai ki manifest me READ_SMS / Notification access / Contacts nahi hain.

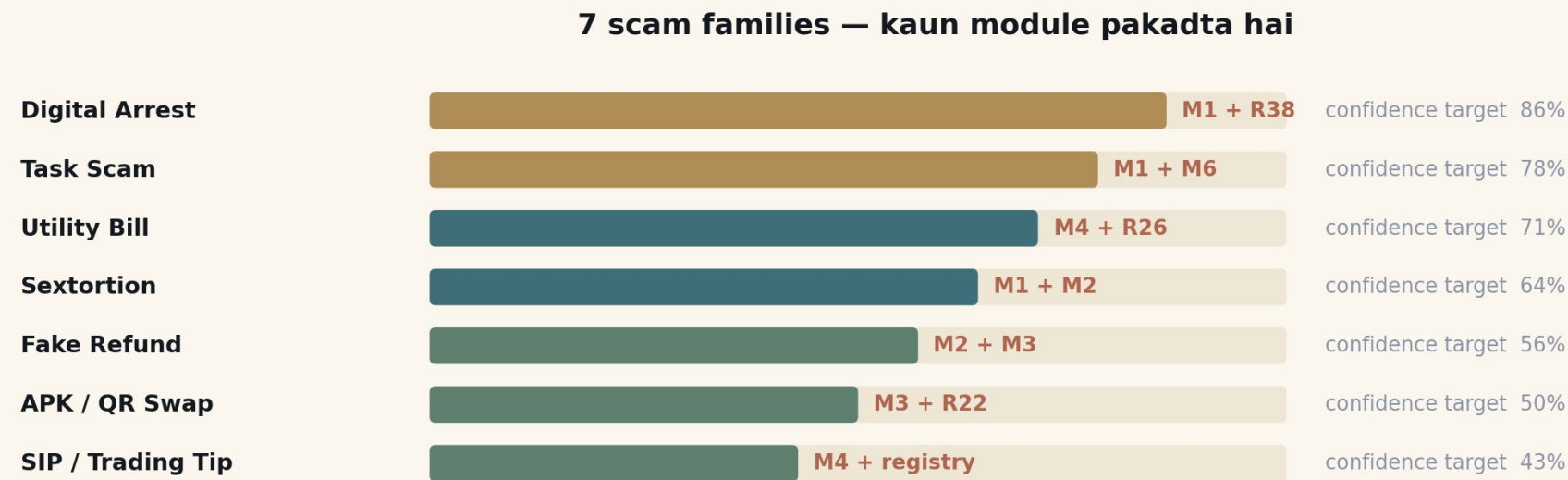
Plaintext message text kahin nahi jaata — server default OFF. Yahi DPDP compliance ka core hai, aur yahi wajah hai ki READ_SMS manifest me nahi hai.

Judges jo score karte hain — humara evidence usi ke against

Resilience & safety impact	30%	M1 pre-pay interception + M7/R38 golden-hour pack — fraud se pehle aur baad, dono
Tier-2/3 usability (Bharat-first)	25%	Hindi voice + senior mode + offline 164KB demo + low-bandwidth text-first UI
Guardrail compliance & trust	15%	0 SMS/notif/contacts permissions · no tips · uncertainty transparent · DPDP-aligned
Technical execution	15%	66/66 checks · P95/R90.5 · 0.012ms median · 33-msg eval harness · CI on every push
Feasibility & scalability	15%	MIT + ~0 per-user cost · Android + web + offline · Phase-2: registry sync, Bhashini, IVR

Har criteria ke liye ek live artifact hai — koi claim bina proof ke nahi.

7 scam families — kaun module pakadta hai



Confidence target = rule design goal (red-team se calibrate) — koi claim nahi; live numbers eval + field me.

Coverage map rule-design se aata hai — har family ke liye kam se kam ek detection path hai, aur ek fallback (human callback / 1930).

Aaj kya nahi hai — aur 30 din me kya hoga

Aaj ki honest limits

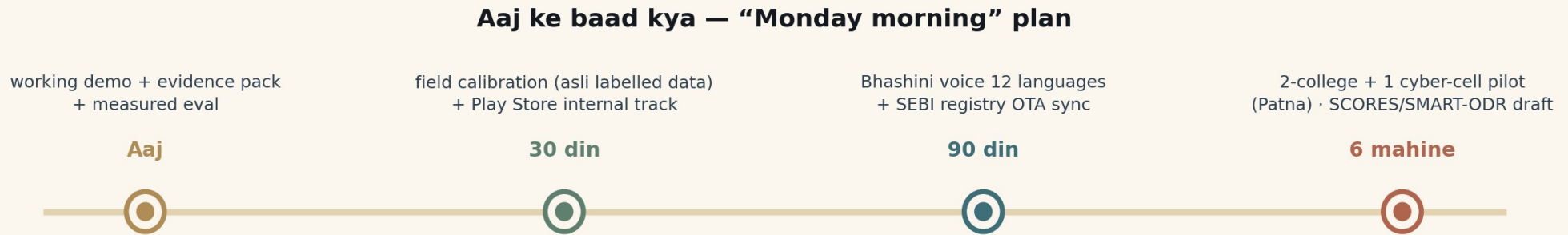
- Eval set self-authored — ceiling hai, field accuracy nahi
- Android telemetry: permission + Play-policy review chahiye
- iOS par kuch signals nahi (graceful degrade)
- OCR opt-in; image store nahi hoti

Phase 2 · 30 din

- Field calibration — asli labelled data par P/R dobara
- Bhashini voice (12 languages) + IVR/missed-call pilot
- SEBI registry live sync + SCORES/SMART-ODR draft
- Pilot: 2 colleges + 1 cyber-cell desk (Patna)

Judges ke liye: jo maine banaya + jo maine jaan-boojh ke nahi banaya — dono likha hua hai (repo me bhi).

Hackathon khatam — phir kya?



Zero-cost rails hi use karenge: GitHub Pages (demo), Play internal track (app), OTA rules JSON (updates).

Asli proof chahiye to hum 30 din me field accuracy dobara measure karke public karenge — ceiling nahi, sach.

Aaj se hi live hai: demo URL, MIT code, 66 tests, measured eval — judge kuch bhi abhi verify kar sakta hai.

Aage ka plan zero-cost rails par hai — koi funding dependency nahi, isliye ye plan realistic hai.

Paisa jaane se pehle rokna — aur fraud ho jaye to 60 minute me complaint.

Live demo kholiye — 2 minute me khud chala kar dekh sakte hain:

sudonishant.github.io/satarkpay-repo

Team SCAMURAI — Nishant Kumar · Prince Singh · Kartik Singh

Code (MIT) · 66/66 checks · P 95% / R 90.5% eval · 0.012 ms median inference · offline demo

Thank you — sawal ke liye taiyaar hain.